
Gamma-Shaped Dealer-Book Pressure and Endogenous Volatility Cycles: A Reduced-Form Fixed-Equilibrium Hopf Model

Mahimn Patel

mahimn.patel@mail.utoronto.ca

Abstract

We construct a reduced-form physical-measure model in which gamma-shaped book pressure interacts with a price–memory–variance loop. Its equilibrium remains fixed as coupling varies and its variance drift points inward at zero. For a centered Gaussian density of signed dealer positions, the local Hopf equation is quadratic in coupling. An illustrative, non-calibrated configuration gives $\kappa^* = 31.4933 \text{ yr}^{-1}$, $\omega^* = 47.1186 \text{ rad yr}^{-1}$, and $\ell_1 = -6.2888$; the actual nonlinear book produces a nearby attracting cycle with positive variance. The kernel shapes a postulated pressure term rather than deriving hedge flow. Public open interest does not identify dealer sign, so the pre-extraction WRDS protocol is participant-sign-free and predictive. No registered market dataset is analyzed.

1 Model and measurement boundary

Option hedging can create state-dependent underlying demand [Frey and Stremme, 1997, Sircar and Papanicolaou, 1998]. Demand-based option pricing [Gârleanu et al., 2009] does not itself derive a spot drift, so our feedback is a reduced-form physical-measure hypothesis. Let $X_t = \log(S_t/F_t)$ for a predictable reference and specify

$$dX_t = [-\delta X_t - \tfrac{1}{2}(v_t - \theta_v) + \kappa g(X_t, v_t, \chi_t)]dt + \sqrt{v_t} dW_t^S, \quad (1)$$

$$dv_t = [\kappa_v(\theta_v - v_t) + \gamma v_t \chi_t]dt + \xi \sqrt{v_t} dW_t^v, \quad (2)$$

$$d\chi_t = \alpha(\beta X_t - \chi_t)dt. \quad (3)$$

At $v = 0$, drift is $\kappa_v \theta_v > 0$ and diffusion is zero. Thus the added feedback is boundary-compatible, although we do not prove global non-explosion.

A Gaussian mass in fixed log moneyness integrates against Black–Scholes gamma as

$$\begin{aligned} \mathcal{G}(X, v) &= \{2\pi\tau^2(v)\}^{-1/2} e^{-q_d T - X} \exp[-(X - m(v))^2/(2\tau^2(v))], \\ \tau^2(v) &= \sigma_q^2 + vT, \quad m(v) = \mu_q - (r - q_d + v/2)T. \end{aligned} \quad (4)$$

We center and normalize it:

$$g(X, v, \chi) = s \frac{\mathcal{G}(X, v) - \mathcal{G}(0, \theta_v)}{\mathcal{G}(0, \theta_v)}, \quad s \in \{-1, +1\}. \quad (5)$$

Then $(0, \theta_v, 0)$ is an equilibrium for every κ . The orientation s describes latent signed dealer positions. Public OI counts contracts but does not identify participant or side [Options Industry Council, 2025]; proprietary participant-tagged data are used when actual market-maker positions are needed [Vasquez et al., 2025].

2 Local bifurcation

At the fixed equilibrium, put $G_X = g_X$, $G_v = g_v$, $G_\chi = g_\chi$. The Jacobian is

$$J = \begin{pmatrix} a & b & d \\ 0 & -\kappa_v & c \\ m & 0 & -\alpha \end{pmatrix}, \quad \begin{aligned} a &= -\delta + \kappa G_X, & b &= -1/2 + \kappa G_v, \\ d &= \kappa G_\chi, & c &= \gamma \theta_v, \end{aligned} \quad m = \alpha\beta. \quad (6)$$

For $\det(\lambda I - J) = \lambda^3 + c_2\lambda^2 + c_1\lambda + c_0$,

$$\begin{aligned} c_2 &= \kappa_v + \alpha - a, & c_1 &= \kappa_v\alpha - a(\kappa_v + \alpha) - dm, \\ c_0 &= -a\kappa_v\alpha - bcm - dm\kappa_v. \end{aligned}$$

Theorem 1. *If the drift is C^3 and, at $\kappa^* > 0$, $c_2, c_1, c_0 > 0$, $H := c_1c_2 - c_0 = 0$, $H' \neq 0$, and $\ell_1 \neq 0$, then the fixed equilibrium undergoes a non-degenerate local Hopf bifurcation. Its eigenvalues are $\pm i\sqrt{c_1}$ and $-c_2$, with*

$$\left. \frac{d \operatorname{Re} \lambda_+}{d\kappa} \right|_{\kappa^*} = -\frac{H'(\kappa^*)}{2(c_1 + c_2^2)}.$$

In the Kuznetsov [2004] convention, $\ell_1 < 0$ gives stable small cycles on the side where the equilibrium pair is unstable.

For (5), $G_\chi = 0$. With $A = \alpha + \kappa_v$, $M = \alpha\kappa_v$, and $L = (\gamma\theta_v)(\alpha\beta)$,

$$H(\kappa) = G_X^2 A \kappa^2 + \{G_v L - G_X A(A + 2\delta)\} \kappa + A(M + \delta A + \delta^2) - L/2. \quad (7)$$

Every root still requires the three positivity conditions and transversality. Negative discriminant means only no local Hopf root. Analytic derivatives of (4) through order three enter the standard first-Lyapunov formula; the variance tensor includes $B_{v,v\chi} = B_{v,\chi v} = \gamma$.

3 Actual-nonlinearity validation

For the illustrative, non-fitted configuration $(\delta, \kappa_v, \theta_v, \alpha, \beta, \gamma) = (0.5, 40, 0.0625, 50, 2, 500)$ and $(\mu_q, \sigma_q, T, s) = (0.06, 0.20, 0.25, +1)$, $G_X = -0.0617978$ and $G_v = -2.0198878$. Enumerating both quadratic roots gives

$$\kappa^* = 31.4932976, \quad \omega^* = 47.1185670, \quad \ell_1 = -6.2888041,$$

with crossing speed 0.2686294. The period is 0.13335 years. The remote second valid root is $\kappa^{**} = 16860.8961$, where the pair crosses back into the stable half-plane; threshold plots use the first root.

Direct integration of (1)–(3) with the actual Gaussian book gives $X \in [-0.02814, 0.02358]$ and $v \in [0.04453, 0.08187]$ after transient discard. The displayed cycle therefore stays in the local, positive-variance region. This validates an internally consistent possibility, not SPX calibration or global stability. Gross-normalized signed mixtures in the full paper show that supercriticality is book-dependent; dispersed or offsetting books can instead be subcritical.

4 Pre-extraction protocol and limitations

Before expected September 2026 WRDS access, Amendments A13–A16 fix the registered historical horizon from 2017-01-03 through 2024-10-29. Four observable OI-and-pricing-derived summaries—log nonnegative mass, call–put composition, mean log moneyness, and log dispersion—separately predict $\log(r_{t+1}^2 + 10^{-8})$ with leads and HAR-style lags constructed on the complete CRSP market calendar, official Cboe VIX, log spot, a linear trend, outcome-session weekday indicators, and one monthly-expiration control. Contracts use fractional settlement time, fixed liquidity/IV/moneyness filters, and parity forwards split by AM/PM settlement; OI is availability-timed. HAC lag 22 and a 22-session complete-calendar bootstrap are separately BH-adjusted over four tests. Robust association requires rejection by both adjusted families and a 95% bootstrap interval excluding zero; one-family rejection is method-sensitive. Convention-signed GEX and stress interactions are secondary. This is a retrospective pre-analysis plan over historically public outcomes, not a blinded prospective experiment.

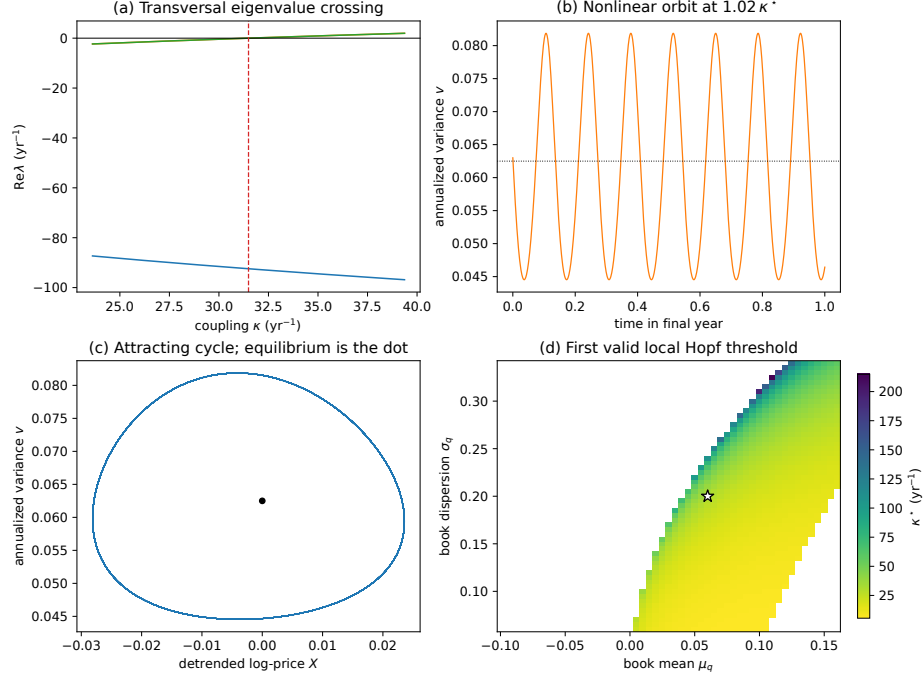


Figure 1: Eigenvalue crossing, actual nonlinear orbit at $1.02\kappa^*$, and local-Hopf map. Blank cells do not imply stability.

The theorem is local, deterministic, and specific to the detrended-log-price coordinate. Brownian paths are not periodic orbits, and no global attractor or stochastic threshold is proved. The pressure term is reduced-form, while the price–memory–variance channel is independently assumed. Any WRDS association concerns public book summaries; confirming signed dealer feedback requires participant-tagged positions and an identification strategy.

References

- Rüdiger Frey and Alexander Stremme. Market volatility and feedback effects from dynamic hedging. *Mathematical Finance*, 7(4):351–374, 1997. doi: 10.1111/1467-9965.00036.
- Nicolae Gârleanu, Lasse Heje Pedersen, and Allen M. Poteshman. Demand-based option pricing. *Review of Financial Studies*, 22(10):4259–4299, 2009. doi: 10.1093/rfs/hhp005.
- Yuri A. Kuznetsov. *Elements of Applied Bifurcation Theory*. Springer, New York, 3rd edition, 2004. doi: 10.1007/978-1-4757-3978-7.
- Options Industry Council. Open interest: Why it matters. Options Education, October 2025. URL <https://www.optionseducation.org/news/open-interest-why-it-matters>. Accessed 9 July 2026.
- K. Ronnie Sircar and George Papanicolaou. General Black–Scholes models accounting for increased market volatility from hedging strategies. *Applied Mathematical Finance*, 5(1):45–82, 1998. doi: 10.1080/135048698334727.
- Aurelio Vasquez, Diego Amaya, Neil D. Pearson, and Pedro Angel Garcia-Ares. 0DTE index options and market volatility: How large is their impact? Technical Report 5113405, SSRN, 2025. URL https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5113405.